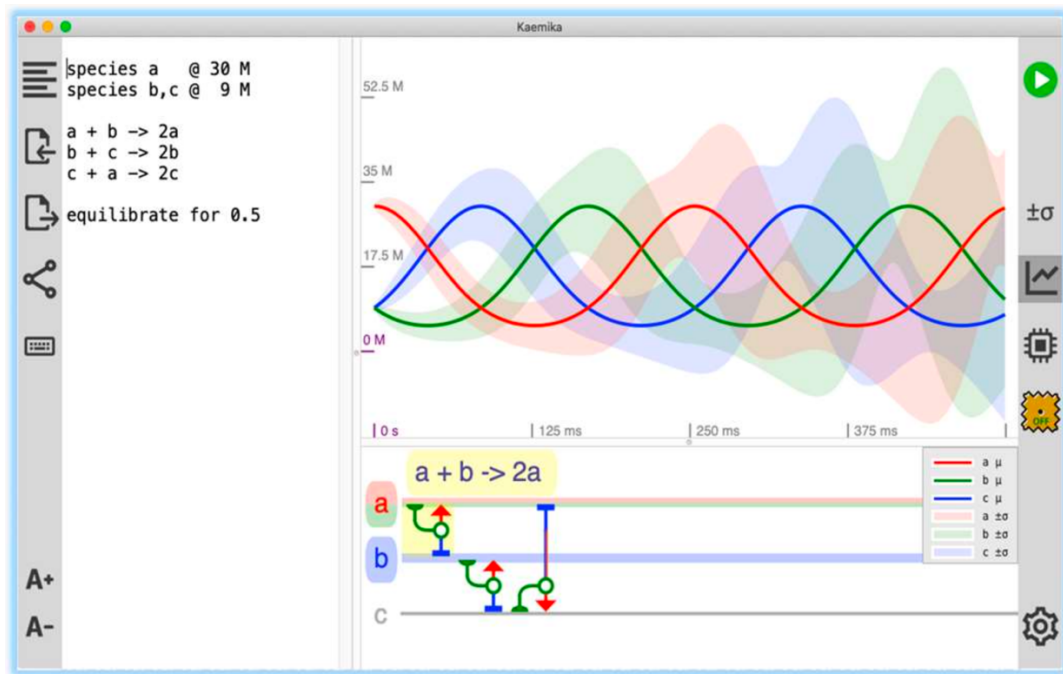




Biocomputing – part 1

AN INTRODUCTION TO KAEMIKA

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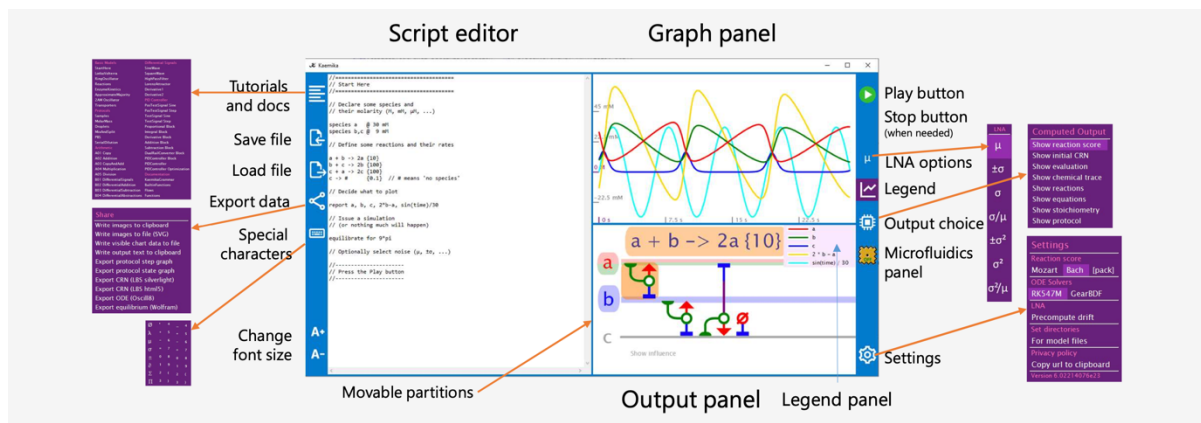
Kaemika developed and distributed by Professor Luca Cardelli, University of Oxford, Oxford, UK

Experiments adopted for 27020 interdisciplinary bioengineering from a tutorial by Professor Geoff Baldwin, Imperial College, London, UK

1. Introduction to Kaemika and a simple $A \rightarrow B$ model

1.1 Kaemika

Kaemika is a chemical reaction simulator, including a modern graphical user interface and a functional programming language. It provides basic chemical and stochastic simulation functionality, supporting natural chemical notation and enhancing it with the recursive and conditional generation of chemical reaction networks.



1.2 Model for $A \rightarrow B$

We are considering a **very simple reaction network**. It takes a **compound 'A'** to be converted into a **compound 'B'**. Following **mass action laws**, the kinetic law associated to the reaction is a **first order reaction**, with **parameter k_1** representing the reaction rate.

Model described: $A \xrightarrow{k_1} B$

Kinetic Law for 'A': $\frac{d[A]}{dt} = -k_1[A]; [A]_{t=0} = A_0 > 0$

Kinetic Law for 'B': $\frac{d[B]}{dt} = k_1[A]; [B]_{t=0} = 0$

Ex1-1: Writing a simple chemical reaction in Kaemika

Kaemika provides a quite elaborate language to describe chemical reactions. In this course, we will only work with a subset of the language. You can code your reaction in a txt-file (which can be loaded into Kaemika) or directly in the Script editor of Kaemika.

Let's express the simple reaction. Compounds are called **species** in Kaemika (we will use the '**bold**' notation to denote keywords of Kaemika in this text),

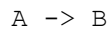
species {A, B}

defines the species, but Just defining the species involved is not enough, we need to tell the amount of species at the beginning. Let's assume that we have 100 mM (milli-mole) of A and nothing of B,

amount A @ 100 mM

amount B @ 0 mM

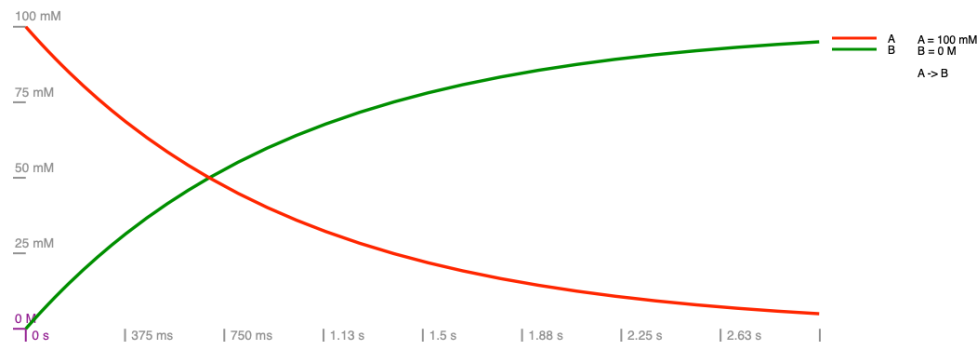
We can now describe the list of reactions that takes place in the sample (Kaemika allows to make different samples, but by default species and reactions are assigned to a sample called "vessel"). In this simple example we only have one reaction,



To start the simulation, we ask it to run for 3 seconds using the **equilibrate** statement,

```
equilibrate for 3
```

Press the 'play' button. You will see the concentrations of A and B over time for 3 seconds in the Graph panel.



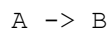
By default, all species are plotted – but by using the keyword **report**, you can specify which species to plot, e.g.,

```
report B
```

will only plot specie B.

Now, select the 'Show initial CRN' (Chemical Reaction Network) from the Output panel. It should show the following,

```
A = 100 mM
B = 0 M
```



```
 $\partial A = - A$ 
 $\partial B = A$ 
```

```
Elapsed Time: 0.007983s
```

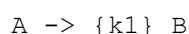
You can recognise the species and their initial concentration as well as the reaction. It also shows the actual mathematical description of the reaction, i.e. the kinetic law, where the direction of mass action follows the arrow, so [A] results in depletion of [A] ($d[A]/dt = - [A]$). This is presented as ODE's (Ordinary Differential Equations) of the chemical reactions, you should be able to recognise them from the kinetic laws for A and B presented above.

Ex1-2: Reaction rates

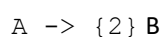
In the first experiment (EX1-1) we did not consider reaction rates (k_1). Kaemika assumes reaction rates are 1 by default, however, we can easily change this. Let's introduce a variable this is done by the keyword '**number**'

```
number k1 = 2
```

We can now express the rate of the reaction transforming A to B,



This is the same at writing,



but using variables comes in handy when the number and complexity of reactions grow.

Task 1: Try different values for k_1 and observe what happens.

NOTE: On some computers, Kaemika has difficulties with decimal numbers! Try to set $k1 = 0.1$, run the simulation at check the CRN output in the lower output panel. If you do not see a 0.1 , try to write the following $k1 = 1/10$. This should work.

Ex1-3: Decay

Having introduced rates, we can now introduce decay, i.e. the fact that species do have a lifetime. Let's define the decay of the produced species B and let's denote the rate of decay by the variable kd . We define a reaction where B turns into zero (expressed as ' $\emptyset$ ' or '#' in Kaemika). We have to add the following to our system,

```
number kd = 0.2
B -> {kd}  $\emptyset$ 
```

Task 2: Run a simulation and explain what you see. Try to change the rates $k1$ and kd and observe.

Extra: Running several samples

As mentioned in Ex1-1, we can explicitly define samples in which to carry out our reactions. To illustrate how this is done in Kaemika, we return to Ex1-1 and define it to run in a sample S1 – you can think of this being a test tube. The Kaemika description is as follows, which the new syntax marked in red,

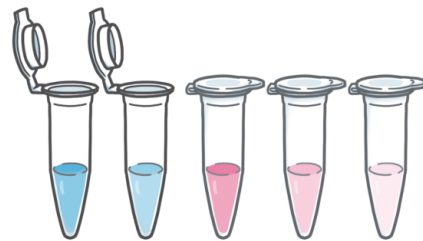
```
sample S1
species {A, B}

amount A @ 100 mM in S1
amount B @ 0 mM in S1

report y1 = A, x1 = B in S1

A -> B

equilibrate S1 for 3
```



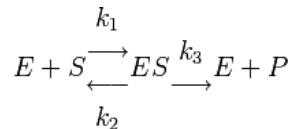
When deciding which species to plot, you can also decide to save the full plot by assigning a variable (name) to the specie, e.g., $y1 = A$ will allow save the curve of A in sample S1 such that you can plot A at a later stage.

2. Modelling enzyme kinetics

In biology, most reactions are catalysed by enzymes. The enzyme has to first bind a substrate so that it can act on it and convert it to products. An enzyme converts a substrate into a product, this is usually an irreversible reaction. An enzyme reaction constitutes a dynamic process and can be studied as such. One may look at the time courses of the reactants or look at the steady-states and their stability properties.

Ex2-1: Enzyme reaction

The enzyme reaction can be expressed as,



where, E is the enzyme, S the substrate, ES the enzyme-substrate complex and P the product. Three kinetic constants define the reaction rates. To make the numbers easier, we will use units of M (M in Kaemika) throughout. Use the following,

$$k_1 = 1; \quad k_2 = 1; \quad k_3 = 0.1$$

k_1 is the diffusion-controlled limit of binding, which is limited by the encounter of molecules in the solution; k_2 is the off rate of the ES complex and k_3 is the reaction rate.

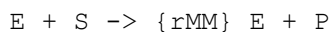
Assume the following initial Conditions: $[E]_{t=0} = 1 \text{ M}$; $[S]_{t=0} = 10 \text{ M}$; $[P]_{t=0} = 0 \text{ M}$;

Task 3: Write the Kaemika code for the enzyme reactions and simulate for 300 sec. Pay special attention to the formation and decay of the [ES] complex. Try changing k_2 to 10, explain the change.

This is a complete model of the reaction scheme that does not rely on the steady state assumption. Each species in the reaction is being accurately modelled.

Ex2-2: Michaelis-Menten approximation (optional)

A convenient approximation, called the **Michaelis-Menten** approximation is often used. This defines the steady state of a reaction as being when $d[ES]/dt = 0$. Using the Michaelis-Menten¹ equation, we can model the enzyme reaction in a much simpler version,



The Michaelis-Menten (overall) reaction rate r_{MM} is a function of the concentration of the substrate S, based on the equation,

$$\frac{d[P]}{dt} = \frac{k_3 * E * S}{K_m + S}$$

where K_m is the Michaelis constant K_M ,

$$K_m = \frac{k_2 + k_3}{k_1}$$

¹ For an explanation of the derivation of the Michaelis-Menten approximation, see https://en.wikipedia.org/wiki/Michaelis-Menten_kinetics.

i.e. $r_{MM} = ((k_3 * E_{tot}) / (K_M + S)) * S$, where E_{tot} is the concentration of E at time 0. In Kaemika we can define this rate as,

```
function rMM (flow S) {  
    (k3*Etot*S) / ((k2+k3) / k1) + S  
}
```

This function can then be used in the reaction as,

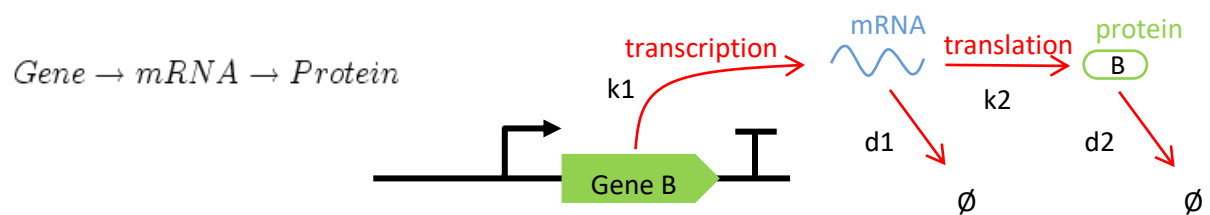
```
E + S -> {rMM(S)} E + P
```

Note that the concentration of a specie is dynamic over time, and in Kaemika declared as the type of `flow`.

Task 4: Write the Kaemika code for the Michaelis-Menten reactions and simulate for 300 sec, using the same parameter values from Task 3. Are there any significant differences between the two kinetic models that you have used?

3. Modelling gene expression

In this section, we explore a computational model to describe a constitutively expressed gene. The model is based on a simple interpretation of the **central dogma**: Gene → (transcription) → mRNA → (translation) → Protein, with both the mRNA molecules and the protein molecules being naturally degraded.



The modelling parameters, used throughout this practical, are characteristic of the E.coli bacteria. The following equations describe the kinetics of the model:

$$\frac{d[mRNA]}{dt} = k_1 - d_1[mRNA]$$
$$\frac{d[Protein]}{dt} = k_2[mRNA] - d_2[Protein]$$

- k_1 is the **transcription rate**. It is considered to be constant, and it represents the number of mRNA molecules produced per gene, and per unit of time. In reality this will be affected by the copy number of the gene, and the number of RNA Polymerases available. Here, we assume these to be constant and so are not included in the model.
- d_1 is the **mRNA degradation rate** of the mRNA molecule. The typical half-life for the mRNAs, in E.Coli, has been measured to be between 2min and 8min (average 5min).
- k_2 is the **translation rate**. It is considered to be constant, and it represents the number of protein molecules produced per mRNA molecule, and per unit of time.
- d_2 is the **protein degradation rate**. In practice the degradation rate of a protein is made of two terms. the first corresponds to break down of the protein. The second is due to dilution of the cell contents when the cell divides. In this practical, we only consider **stable proteins**, with no active degradation pathways. We can therefore approximate the degradation of the protein by the dilution term.

Ex3-1: Constitutive gene expression

Task 5. Define the reaction model in Kaemika and calculate the rates k_1 , k_2 , d_1 , and d_2 .

Note: both the gene and the mRNA are modifiers of the transcription and translation reactions. It is important that they are not included as Reactants, because they would be 'consumed' during the reaction. Both genes and mRNA act as templates and can produce more than one molecule of product and are themselves not converted into the products – mRNA and protein.

The formation of mRNA is only dependent on the copy number of the gene and the rate of transcription. Concentration of nucleotides and RNA Polymerase are not included in the expression; this assumes that these species are not limiting for transcription. This is generally true at low levels of transcription from a single gene, but this assumption may not be valid with high copy numbers, limiting nutrients, strong promoters etc.

Now let us define the simulation constants. Keep your units constant, the values given are in minutes, so this would be a sensible unit to use throughout.

Q1. d1: the average half-life of mRNA is 5 minutes. What is the rate of decay?

Half-life and rate have an exponential relationship (an example most people are familiar with is the half-life of radioactive decay). From the maths of this relationship it can be shown that:

$$rate(k) = \frac{\ln 2}{\text{half life } (t_{1/2})}$$

If you want to understand the maths, consult a physical chemistry textbook.

Q2. d2: what is the dilution rate during exponential E. coli growth?

It can be assumed that a bacterium divides once its volume has doubled. Knowing that the typical cell division time for e-coli is 30-40min (use 35 minutes), what is the dilution rate?

Q3: k₁: Knowing that average number of mRNA molecules per gene is 2.5 in E.Coli, what is the average transcription rate ?

You should consider the steady state situation where [mRNA]=2.5

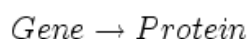
Q 4: k₂: given that the average number of proteins per gene is 1000 in E.Coli, what is the average translation rate ?

You are now ready to run the simulations with Kaemika. Define all the necessary kinetics laws for the model and create all the appropriate parameters. Vary the timescale and only plot the important parameters (you can ignore nucleotides, amino acids, gene, etc).

Task 6: Run a simulation using the parameters you have calculated above. [Check the final values of mRNA and protein, are they correct?]

Ex3-2: Simplified gene expression

The above simulation has a number of assumptions. Depending on the situation, it may be appropriate to simplify the simulation even further and not worry about [mRNA]. The model below demonstrates this.



The simulation now reduces to this simplified situation:

$$\frac{d[Protein]}{dt} = k - d[Protein]$$

Task 7. Run a simulation using the same protein synthesis and degradation parameters as in task 2. (Note that the translation rate should take into account the steady state [mRNA] from your previous simulation).

What is the key difference between the simplified and full simulations (consider the approach to steady state).